Islamic Finance: How Relevant is Debt in Indian Companies?

by:

Moid U. Ahmad

Assistant Professor,

Jaipuria Institute of Management,

A-32A, Sector 62, Noida,

UP-201309, India

(M):9717720779

Email: professor.moid@gmail.com

Islamic Finance: How Relevant is Debt in Indian Companies?

Abstract:

Any concept based on ethics should also have economic viability and should be logical. Islamic financial system is one such concept which requires a lot of effort to bring economic viability and technical expertise into its ambit. This study aims to provide one such technical perspective of the system. This is an empirical attempt to study financial risk in companies from the perspective of Islamic finance. The basic philosophy of the study is that, Debt is not advisable and should be avoided by companies. It tries to analyze the relationship between Debt, Equity and Profitability of 'Sensex' companies in India. It hopes to make a significant contribution to the available literature on the subject, especially in economic scenario such as India.

Keywords: Islamic economics, Interest free financing, corporate finance, Debt

JEL Classification: C2, G3

Introduction:

'Capitalism' versus 'Socialism' has been a constant debate ever since the two concepts evolved. Wealth has always been an issue of extreme importance to every faith and is responsible for many evils in the society. There is a continuous discussion going on issues related to Islamic economics worldwide, which is an encouraging sign. The Muslims are discussing the issues not just for their welfare but the welfare of the whole mankind. When we talk of resilience of Islamic Financial System to deal with financial crisis, as the current one, it becomes even more imperative to focus on its acceptability and generate a consensus on its utility. In a society with people from diverse faith it is only possible when at least the issues of common interest are discussed and implemented. Movements reflecting such ideas have been a part of philosophy and belief of major religions across the globe since time immemorial. One of the religions which offer a logical and comprehensive view on the subject is, Islam as it offers a complete guide for the well being of human being in this life as well as life after death.

Interest is perhaps the root cause of the all evils in the world. It may be the 'economic interest' or the 'vested interest'. They are interlinked and interdependent and prohibited not only in Islam but also in Judaism and Christianity (Jafri and Margolis, 1999).This interest is also the building block of the modern day financial system. Thus a system based on a concept which is not 'just' cannot do justice to its beneficiaries. Hence, the need for an alternative economic system. The question of Islamic economy being a social economy rather than a viable economy has been raised time over time. What exactly is the objective of Islamic economics is an issue of contention. Timur Kuran (1997) writes that 'the declared purpose of Islamic economics is to identify and establish an economic order that conforms to Islamic scripture and traditions'.

The concept of Islamic economic has been there since there has been Islam,infact, long before that, in principle. Almost four decades have passed since the first Islamic bank began its operations in Egypt, and more than a decade has passed since the Islamic Republics of Iran and Pakistan adopted a non-interest-based financial system. The growth of financial institutions, instruments, and transactions operating without interest has been impressive. Today, Islamic banking is spreading and gaining acceptance in non-Muslim countries as well as Muslim ones. Equally important has been the growth of scholarly interest in the subject. The viability and feasibility of noninterest-based financial transactions, instruments is been proved and accepted globally. According to senior Islamic scholar Prof. Umer Chapra (2000), in one of his studies comments about research in this area, 'substantial but still far from adequate literature is available is Islamic finance'.

The Profit Loss Sharing model Islamic banking developed by Siddiqi (1967) the Islamic economy was visualized as a "debt-free economy".The capital structure of companies consists of various debt and equity constituents. Debt is prohibited in Islam as it involves interest, a form of 'Riba', but equity contracts are permitted selectively. The rationale behind the study is to compare the performance of companies with various levels of debts in their capital structure and draw conclusions to bolster the case of equity participation against interest based financing. Shariah refers to Islamic law which is based on the teachings of the Quran and Halal products refer to the products that fulfill the criteria laid out by Islam of being acceptable for use.

Literature Review:

The combination of contracts is a potential mechanism of product development in Islamic finance. However, this concept encounters legal issues due to a Hadith that prohibit two contracts in one deal. Burhan (2007) argues for the validity of combining two or more contracts to structure Shariah-compliant products. It discusses many aspects of combination of contracts, including terminologies and purposes of the contracts, the degree of uncertainty and ambiguity, and the nature of the bargain in the contracts combined.

Ahmad et al. (2006), in their article investigates the efficiency of the full-fledged Islamic banks, Islamic windows and conventional banks using a stochastic frontier approach (SFA) in Malaysia. It analyzed the technical and cost efficiency of SFA on Islamic banking industries in terms of assets, deposits and financing base which grew rapidly between 1997 and 2003. Looney (1982) verified that Islam and economic growth can move forward simultaneously by taking up the example of Saudi Arabia.

‘A unique feature of Islamic banking, in theory, is its profit-and-loss sharing (PLS) paradigm’, Chong and Liu (2009). There study was one of its kind and such types of studies are required to progress with the deliberations on Islamic Finance. In practice, however, we find that Islamic banking is not very different from conventional banking. Their study on Malaysia shows that only a negligible portion of Islamic bank financing is strictly PLS based and that Islamic deposits are not interest-free, but is closely pegged to conventional deposits. They concluded that the rapid growth in Islamic banking is largely driven by the Islamic resurgence worldwide rather than by the advantages of the PLS paradigm and that Islamic banks should be subject to regulations similar to those of their western counterparts.

Jobst et al.(2008) reviewed the key developments in the Sukuk market and debates about challenges and opportunities going forward. The paper finds that while the Sukuk market continues to generate strong interest by new issuers in Muslim and non-Muslim countries alike, some critical constraints arising from continued legal uncertainty and regulatory divergences still need to be overcome for an Islamic Bond market to flourish. Alanazi et al. (2011) studied Saudi Arabian initial public offerings' (IPOs) financial performance before and after going public on the Saudi Stock Exchange Market and found out that Saudi IPOs exhibit a significant decline in the post-IPO performance compared to the pre-IPO level as measured by the return on assets and return on sales. It was also found that the performance deterioration is associated with the IPO event. Shubber and Alzafiri (2008) studied the computation of cost of capital for Islamic banks as for Islamic banks, it became clear that the deposit accounts were not a liability, as these fell within the definition of “profit-and-loss sharing” instruments. A high-positive correlation coefficient was found between an Islamic bank's market value and the size of its deposits. Also, the market value of Islamic banks was found to be independent of its cost of capital. Ahmad and Rehman (2012) compared the efficiency of Islamic and Conventional commercial banks in Malaysia and found out that there is not much significant difference in the performance of both types of banks and later outperformed the former on all efficiency parameters.

Rosly and Bakar (2003) found that Islamic banking scheme (IBS) banks have recorded higher return on assets (ROA) as they are able to utilize existing overheads carried by mainstream banks. As this lowers their overhead expenses, it is found that the higher ROA ratio for IBS banks does not imply efficiency. It is also inconsistent with their relatively low asset utilization and investment margin ratios.

Farhang (2009) emphasizes on the concept of capital and ownership of capital in market economy and Islamic economy. He states that the role capital is expected to play in the economy is different in two economic systems. This paper reviews the concept of capital and the process of capital formation both before and after industrial revelation. In that context it also looks at the function of interest rate during the time of Prophet Mohammed and the present time. Mondher (2003) discusses the short and long-term financing and capital structure of Islamic firms. He also discuses certain examples of long and short term Islamic debt instruments used in GCC and Malaysia. He also discusses the modern capital structure theories (Modigliani-Miller approach) to study its utility in Islamic finance. He concludes that the concept of financial risk and is not adequate to the capital structure of capital of an Islamic bank. Moreover under certain assumptions, the increase in investment account financing increases the return of shareholders without an extra risk and increases the market value of Islamic bank with WACC remaining constant. He also gives a model to calculate Weighted Average Cost of Capital (WACC) for a bank. According to him the sources of funds for an Islamic bank are Shareholders' equity and reserves, Current account, Investment accounts and Savings accounts. Esty (2000) focuses on the development of project finance structures for Islamic countries using illustrations from a case study of Equate Petrochemical Company, a petrochemical plant in Kuwait. Modi (2007) reflects on the need for standardizing the regulation of Islamic finance. He cites that a regulatory framework that would ensure transparency and integration is critical. He believes that governments could consider developing such framework to capture a share in the Islamic finance market. Bashir et al. (1993), present a two-period general equilibrium model of investment under uncertainty and symmetric information. An optimal solution for the investment model is also derived for a representative Islamic bank. It is theoretically shown that equity capital and the profit-sharing ratios are among the primary determinants of investment, each of which exerts a positive effect on investment. Besides these two factors, potential effects from depreciation and the rate of inflation on the level of investment are also considered in the theoretical analysis. An alternative theoretical model is also derived for the same.

Research Methodology:

The primary objective of the study is to understand the effect of Debt and Equity in a company upon its profitability. Some of the imperative research questions which the study aims to answer are :

1. Does a company with zero debt perform better than a company with debt?
2. Does a company with less debt perform better than company with more debt?
3. How do debt and equity combined affects PAT and how do they affect each other?
4. What is the relationship between debt ratio and PAT?
5. How does interest payments affects the profitability of a company?

To study the implications of the concepts of Islamic finance, a good idea is to apply the concepts on companies operating in an Interest based economy. A similar research was done by Ahmad (2012) on Midcap companies in India. Thus, the focus of the study is the capital structure of companies operating in the Republic of India. Sensitive Index (SENSEX) is the most important index of Bombay Stock Exchange in India, which consists of 30 blue chip companies and for the purpose is taken as a representative sample of Indian economy. Data used in the study has been taken from PROWESS, the database maintained by Centre for Monitoring of Indian Economy (CMIE), which is considered to be very authentic. Some of the significant parameters used for study are :

a) Debt
b) Equity
c) Debt ratio as Debt/(Debt + Equity)
d) Profit after Tax(PAT)
e) Interest Paid(IP)
f) Profit Before Interest and Taxes(PBIT)

Equity used here is the amount of paid up equity capital which was adjusted from capital employed to calculate the debt as:

Debt = Capital employed – Equity

PAT has been regressed over debt and equity for all the companies of the sample to study the relationship of capital structure with its profitability. Also correlation between debt ratios with PAT for each company has been calculated to study the relation of debt with PAT. The ratio of Interest Paid over PBIT (The inverse of Interest Coverage ratio) has been calculated which indicates the payments made as interest out of the operating profit, an expense which can be avoided if there is no debt in the capital structure. The time period taken is ten years (2001-2010) and for the 30 SENSEX companies as on March 31, 2010. For some companies the data is available only till March 2009.

The sample used is illustrated in table 1.

Table 1: List of Sample companies, time period of data and acronyms for companies

Company	No. of Years	Acronyms
A C C Ltd.	10	ACC
Bharat Heavy Electricals Ltd.	9	BHEL
Bharti Airtel Ltd.	9	Bharti
Cipla Ltd.	9	Cipla
D L F Ltd.	9	DLF
H D F C Bank Ltd.	10	HDFCB
Hero Honda Motors Ltd.	9	HH
Hindalco Industries Ltd.	9	Hindalco
Hindustan Unilever Ltd.	9	HUL
HDFC Ltd.	10	HDFCL
I C I C I Bank Ltd.	10	ICICI
I T C Ltd.	10	ITC
Infosys Technologies Ltd.	10	Infy
Jaiprakash Associates Ltd.	9	JPA
Jindal Steel & Power Ltd.	9	JSPL
Larsen & Toubro Ltd.	9	L&T
Mahindra & Mahindra Ltd.	10	MM
Maruti Suzuki India Ltd.	9	MSIL
N T P C Ltd.	9	NTPC
Oil & Natural Gas Corpn. Ltd.	9	ONGC
Reliance Communications Ltd.	5 (2005-2009)	Rcom
Reliance Industries Ltd.	10	RIL
Reliance Infrastructure Ltd.	9	Rel Infra
State Bank Of India	10	SBI
Sterlite Industries (India) Ltd.	10	Sterlite
Tata Consultancy Services Ltd.	10	TCS
Tata Motors Ltd.	9	TATAM
Tata Power Co. Ltd.	9	TATAP
Tata Steel Ltd.	9	TATAS
Wipro Ltd.	10	Wipro

Data Analysis:

The sample has been defined in Table 1. Other tables represent the data analysis to draw conclusion in order to answer the research questions. In Table 2, names with suffix 'E' indicates figures for equity, 'dr' indicate Debt Ratio and 'PAT' indicates Profit After Tax figures while names with no suffix indicate Debt in companies.

Table 2: Basic statistics for Debt, Equity , PAT and Debt Ratio for Sensex companies

	Range	Mean	S.D.
ACC	1181.26	1.0017E3	439.26
BHEL	761.58	4.4465E2	271.49
BHEL.PAT	2825.60	1.4364E3	1115.82
ACC.PAT	1549.56	6.9042E2	613.10
Bharti	7597.34	3.0603E3	2965.09
Bharti.PAT	2.60E4	9.7055E3	8690.52
Cipla	775.26	2.5068E2	253.27
Cipla.PAT	597.74	4.5484E2	234.71
DLF	8693.72	2.8123E3	3397.98
DLF.PAT	2548.92	5.5309E2	907.62
HDFCB	1.12E4	4.8510E3	3767.08
HDFCB.PAT	2738.58	1.0866E3	913.06
HH	141.24	1.3870E2	48.50
HH.PAT	1034.89	7.6758E2	307.49
Hindalco	7071.48	4.1753E3	2810.86
Hindalco.PAT	2278.80	1.4917E3	879.52

HUL	1496.74	4.1774E2	606.54
HUL.PAT	1299.11	1.7164E3	388.61
HDFCL	7.46E4	4.4730E4	26150.64
HDFCL.PAT	2367.95	1.4034E3	847.99
ICICI	9.13E4	5.5917E4	28643.15
ICICI.PAT	3996.63	2.2859E3	1480.65
ITC	652.66	2.3148E2	191.89
ITC.PAT	3055.34	2.2736E3	1008.29
Infy	.00	.0000	.00000
Infy.PAT	5190.19	2.7837E3	2036.54
JPA	1.19E4	4.0065E3	3871.21
JPA.PAT	926.94	3.3909E2	320.69
JSPL	4235.66	2.0029E3	1493.15
JSPL.PAT	1435.23	5.8049E2	510.23
ACC.E	5227.25	2.5775E3	1883.30
BHEL.E	9919.20	6.9921E3	3280.04
Bharti.E	2.55E4	9.0969E3	8568.05
Cipla.E	2921.18	1.8444E3	1159.78
DLF.E	1.10E4	2.5372E3	4353.45
HDFCB.E	1.99E4	6.5037E3	6629.15
Hindalco.E	1.76E4	9.8192E3	5889.77
HH.E	3194.64	1.7834E3	1119.78
HUL.E	2277.84	2.3350E3	695.59
HDFCL.E	1.18E4	6.1732E3	4528.44

ICICI.E	4.99E4	2.2493E4	19258.26
ITC.E	1.10E4	8.5307E3	3882.82
Infy.E	2.06E4	8.6220E3	7195.69
JPA.E	5492.78	1.7506E3	1749.26
JSPL.E	4514.18	1.9345E3	1595.36
ACC.dr	.55	.3590	.23
BHEL.dr	.21	.0811	.06
Bharti.dr	.53	.2056	.18
Cipla.dr	.16	.1044	.06
DLF.dr	.84	.4867	.28
HDFCB.dr	.23	.4800	.07
Hindalco.dr	.23	.2767	.07
HH.dr	.13	.0944	.04
HUL.dr	.42	.1356	.17
HDFCL.dr	.06	.8800	.02
ICICI.dr	.42	.7200	.12
ITC.dr	.19	.0410	.05
Infy.dr	.00	.0000	.0000
JPA.dr	.21	.6589	.059
JSPL.dr	.14	.5433	.051

Table 2: Contd…

	Range	Mean	S.D.
NTPC	2.49E4	1.9326E4	8159.34
NTPC.PAT	4661.70	5.5833E3	1716.15
ONGC	1.08E4	2.4348E3	3545.33
ONGC.PAT	1.15E4	1.1834E4	4356.50
RCom	2.88E4	1.2186E4	12316.35
RCom.PAT	4802.67	1.9607E3	2020.87
RIL	6.11E4	2.9839E4	20288.38
RIL.PAT	1.69E4	9.4766E3	5994.45
Rel Infra	6260.55	3.2317E3	2351.07
Rel Infra.PAT	1017.01	5.7294E2	372.01
SBI	8.52E4	4.2016E4	31986.86
SBI.PAT	7561.80	4.9091E3	2619.42
Sterlite	3689.31	2.3853E3	1067.19
Sterlite.PAT	1145.39	5.0978E2	412.54
TCS	375.03	70.9611	123.09
TCS.PAT	5617.49	2.5719E3	2215.66
TATAM	8307.54	3.3773E3	2595.49
TATAM.PAT	2529.26	9.2013E2	870.27
TATAP	3197.58	2.9485E3	874.96
TATAP.PAT	522.50	6.2464E2	173.57
TATAS	2.45E4	8.4429E3	8512.57
TATAS.PAT	4996.84	2.7343E3	1882.62

Wipro	4581.11	1.2579E3	1913.64
Wipro.PAT	4240.58	2.0544E3	1377.86
MSIL	1002.06	6.2533E2	298.19
MSIL.PAT	2000.20	7.8642E2	697.85
MM	3315.49	1.7124E3	1080.24
MM.PAT	1990.84	7.1736E2	617.97
L&T	5039.48	2.8021E3	1564.48
L&T.PAT	3166.55	1.1867E3	1049.94
NTPC.E	3.40E4	4.1379E4	11757.81
ONGC.E	4.90E4	4.8769E4	17874.47
Rcom.E	4.79E4	2.0941E4	17362.55
RIL.E	1.13E5	5.3687E4	38157.07
Rel Infra.E	6605.79	5.4131E3	2768.97
SBI.E	5.16E4	3.1011E4	18286.75
Sterlite.E	1.88E4	5.9142E3	6184.57
TCS.E	1.51E4	6.2806E3	5945.02
TATAM.E	6940.91	4.4805E3	2331.69
TATAS.E	2.69E4	1.1252E4	10407.52
Wipro.E	1.30E4	6.6341E3	4236.21
MSIL.E	7138.91	4.8916E3	2574.23
MM.E	6394.66	3.1719E3	2080.82
L&T.E	9328.81	4.9711E3	3200.068
TATAP.E	4365.16	5.5718E3	1526.28
NTPC.dr	.09	.3111	.03

ONGC.dr	.22	.0589	.077
RCom.dr	.45	.2500	.22
RIL.dr	.15	.3710	.05
Rel Infra.dr	.26	.3300	.10
SBI.dr	.20	.5420	.07
Sterlite.dr	.44	.3860	.15
TCS.dr	.89	.1056	.29
TATAM.dr	.30	.4122	.09
TATAS.dr	.37	.4356	.12
Wipro.dr	.28	.0910	.11
MSIL.dr	.28	.1333	.08
MM.dr	.24	.3590	.07
L&T.dr	.28	.3711	.10
TATAP.dr	.15	.3467	.05

Table 3: Correlations between PAT and Debt Ratio for the sample

* indicates Correlation is significant at the 0.05 level

** indicates Correlation is significant at the 0.01 level

For Infosys Debt ratio is zero for the time period.

Company	**Correlations**
A C C	-0.977 **
BHEL	-0.86 **
Bharti	-0.758 *
Cipla	0.537
D L F	0.008

HDFCB	-0.873 **
HH	-0.744 *
Hindalco	0.657
HUL	-0.115
HDFCL	-0.439
ICICI	-0.274
ITC	-0.584
Infy	NA
JPA	0.413
JSPL	-0.356
L&T	-0.485
MM	-0.552
MSIL	-0.78 *
NTPC	0.868 **
ONGC	-0.782 *
RCom	0.835
RIL	-0.705 *
Rel Infra	0.717
SBI	0.563
Sterlite	-0.825 **
TCS	-0.439
TATAM	-0.427
TATAP	-0.177
TATAS	-0.595
Wipro	0.769 **

Table 4: Regression of PAT on Debt and Equity for Sample companies

Company	Regression Equation	R Squared
A C C	PAT= -0.578 * DEBT + 0.189 * EQUITY + 782.44	0.94
BHEL	PAT= -0.49 * DEBT + 0.29 * EQUITY -412.45	0.96
Bharti	PAT= 0.20 * DEBT + 0.951 * EQUITY + 439.467	0.99
Cipla	PAT= 0.213 * DEBT + 0.155 * EQUITY + 116.38	0.90
DLF	PAT= 0.041 * DEBT + 0.164 * EQUITY + 22.23	0.85
HDFCB	PAT= 0.119 * DEBT + 0.070 * EQUITY + 53.46	0.994
HH	PAT= 1.82 * DEBT + 0.26 * EQUITY + 50.81	0.94
Hindalco	PAT= 0.381 * DEBT- 0.042 * EQUITY + 311.56	0.92
HUL	PAT= -0.206 * DEBT- 0.17 * EQUITY + 2207.82	0.13
HDFCL	PAT= 0.020 * DEBT + 0.071 * EQUITY + 62.78	0.98
ICICI	PAT= 0.001 * DEBT + 0.70 * EQUITY + 637.72	0.87
ITC	PAT= 0.472 * DEBT + 0.267 * EQUITY - 115.08	0.96
Infy	Since Debt is zero ,equation was not derived	
JPA	PAT= -0.132 * DEBT + 0.45 * EQUITY + 10.62	0.89
JSPL	PAT= -0.0006 * DEBT + 0.238 * EQUITY + 0.167	0.96
L&T	PAT= -0.11 * DEBT + 0.36 * EQUITY - 289.56	0.95
MM	PAT= -0.268 * DEBT + 0.39 * EQUITY - 62.45	0.95
MSIL	PAT= -0.37 * DEBT + 0.25 * EQUITY - 205.51	0.80
NTPC	PAT= -0.005 * DEBT + 0.147 * EQUITY - 397.02	0.96
ONGC	PAT= -0.233 * DEBT + 0.198 * EQUITY + 2743.87	0.87
RCom	PAT= 0.148 * DEBT + 0.011 * EQUITY - 95.27	0.996
RIL	PAT= -0.28 * DEBT + 0.28 * EQUITY + 2887	0.81
Rel Infra	PAT = 0.049 * DEBT + 0.088 * EQUITY - 63.74	0.95
SBI	PAT= 0.22 * DEBT - 0.045 * EQUITY - 1.08	0.995
Sterlite	PAT= -0.032 * DEBT + 0.055 * EQUITY + 260.48	0.564
TCS	PAT= 0.368 * DEBT + 0.166 * EQUITY + 246	0.96
TATAM	PAT= -0.47 * DEBT + 0.73 * EQUITY - 750	0.90
TATAP	PAT= 0.027 * DEBT + 0.10 * EQUITY - 17.61	0.97
TATAS	PAT= -0.23 * DEBT + 0.33 * EQUITY + 944.10	0.84
Wipro	PAT= -0.038 * DEBT + 0.33 * EQUITY - 143.41	0.989

Table 5: Interest Paid to Operating Profit (IP/PBIT) ratio for Sample companies

Year	Company	IP/PBIT	Company	IP/PBIT	Company	IP/PBIT	Company	IP/PBIT
Mar-01	A C C	0.764	BHEL	0.123	Bharti	0.878	Cipla	0.003
Mar-02	A C C	0.507	BHEL	0.127	Bharti	0.985	Cipla	0.007
Mar-03	A C C	0.550	BHEL	0.069	Bharti	0.991	Cipla	0.005
Mar-04	A C C	0.311	BHEL	0.052	Bharti	0.982	Cipla	0.026
Mar-05	A C C	0.185	BHEL	0.049	Bharti	0.139	Cipla	0.015
Mar-06	A C C	0.094	BHEL	0.022	Bharti	0.077	Cipla	0.016
Mar-07	A C C	0.048	BHEL	0.012	Bharti	0.043	Cipla	0.009
Mar-08	A C C	0.052	BHEL	0.008	Bharti	0.029	Cipla	0.013
Mar-09	A C C	0.046	BHEL	0.006	Bharti	0.025	Cipla	0.035
Mar-10	A C C	0.054						
Year	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**
Mar-01	HDFCB	0.705	HH	0.007	Hindalco	0.075	HUL	0.008
Mar-02	HDFCB	0.716	HH	0.002	Hindalco	0.082	HUL	0.004
Mar-03	HDFCB	0.676	HH	0.002	Hindalco	0.165	HUL	0.004
Mar-04	HDFCB	0.627	HH	0.002	Hindalco	0.130	HUL	0.029
Mar-05	HDFCB	0.573	HH	0.002	Hindalco	0.092	HUL	0.079
Mar-06	HDFCB	0.606	HH	0.002	Hindalco	0.087	HUL	0.011
Mar-07	HDFCB	0.660	HH	0.001	Hindalco	0.078	HUL	0.005
Mar-08	HDFCB	0.667	HH	0.001	Hindalco	0.143	HUL	0.011
Mar-09	HDFCB	0.713	HH	0.001	Hindalco	0.177	HUL	0.008
Mar-10	HDFCB	0.625						
Year	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**
Mar-01	ICICI	0.787	I T C	0.055	Infy	0.000	JPA	1.466

Mar-02	ICICI	0.843	I T C	0.038	Infy	0.000	JPA	0.833
Mar-03	ICICI	0.903	I T C	0.017	Infy	0.000	JPA	0.464
Mar-04	ICICI	0.783	I T C	0.012	Infy	0.000	JPA	0.433
Mar-05	ICICI	0.722	I T C	0.015	Infy	0.000	JPA	0.394
Mar-06	ICICI	0.756	I T C	0.005	Infy	0.000	JPA	0.239
Mar-07	ICICI	0.818	I T C	0.002	Infy	0.000	JPA	0.293
Mar-08	ICICI	0.823	I T C	0.004	Infy	0.000	JPA	0.287
Mar-09	ICICI	0.816	I T C	0.006	Infy	0.000	JPA	0.286
Mar-10	ICICI	0.767	I T C	0.012	Infy	0.000		

Table 5: Contd…

Year	Company	IP/PBIT	Company	IP/PBIT	Company	IP/PBIT	Company	IP/PBIT
Mar-01	D L F	0.541	L&T	0.580	MM	0.491	MSIL	-0.354
Mar-02	D L F	0.339	L&T	0.484	MM	0.572	MSIL	0.400
Mar-03	D L F	0.113	L&T	0.302	MM	0.375	MSIL	0.162
Mar-04	D L F	0.156	L&T	0.102	MM	0.152	MSIL	0.052
Mar-05	D L F	0.236	L&T	0.064	MM	0.041	MSIL	0.027
Mar-06	D L F	0.280	L&T	0.087	MM	0.024	MSIL	0.012
Mar-07	D L F	0.445	L&T	0.045	MM	0.014	MSIL	0.021
Mar-08	D L F	0.200	L&T	0.039	MM	0.059	MSIL	0.023
Mar-09	D L F	0.405	L&T	0.071	MM	0.126	MSIL	0.030
Mar-10					MM	0.061		

Year	Company	IP/PBIT	Company	IP/PBIT	Company	IP/PBIT	Company	IP/PBIT
Mar-01	HDFCL	0.750	RCom	Loss	RIL	0.304	Rel Infra	0.146
Mar-02	HDFCL	0.725	RCom	0.000	RIL	0.292	Rel Infra	0.162
Mar-03	HDFCL	0.694	RCom	0.173	RIL	0.238	Rel Infra	0.337
Mar-04	HDFCL	0.641	RCom	0.246	RIL	0.186	Rel Infra	0.162
Mar-05	HDFCL	0.605	RCom	0.167	RIL	0.139	Rel Infra	0.141
Mar-06	HDFCL	0.613			RIL	0.076	Rel Infra	0.191
Mar-07	HDFCL	0.649			RIL	0.076	Rel Infra	0.188
Mar-08	HDFCL	0.604			RIL	0.045	Rel Infra	0.200
Mar-09	HDFCL	0.688			RIL	0.086	Rel Infra	0.201
Mar-10	HDFCL	0.632			RIL	0.088		
Year	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**
Mar-01	JSPL	0.263	TCS	0.000	TATAM	-2771.722	TATAP	0.348
Mar-02	JSPL	0.304	TCS	0.000	TATAM	1.153	TATAP	0.321
Mar-03	JSPL	0.293	TCS	0.012	TATAM	0.306	TATAP	0.300
Mar-04	JSPL	0.146	TCS	0.005	TATAM	0.069	TATAP	0.262
Mar-05	JSPL	0.093	TCS	0.001	TATAM	0.046	TATAP	0.196
Mar-06	JSPL	0.115	TCS	0.001	TATAM	0.060	TATAP	0.181
Mar-07	JSPL	0.144	TCS	0.001	TATAM	0.077	TATAP	0.206
Mar-08	JSPL	0.130	TCS	0.001	TATAM	0.102	TATAP	0.182
Mar-09	JSPL	0.104	TCS	0.001	TATAM	0.424	TATAP	0.270

Table 5: Contd…

Year	Company	IP/PBIT	Company	IP/PBIT
Mar-01	N T P C	0.155	ONGC	0.037
Mar-02	N T P C	0.167	ONGC	0.021
Mar-03	N T P C	0.201	ONGC	0.003
Mar-04	N T P C	0.183	ONGC	0.003
Mar-05	N T P C	0.125	ONGC	0.002
Mar-06	N T P C	0.141	ONGC	0.002
Mar-07	N T P C	0.165	ONGC	0.001
Mar-08	N T P C	0.153	ONGC	0.002
Mar-09	N T P C	0.171	ONGC	0.005
Year	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**
Mar-01	SBI	0.873	Sterlite	0.457
Mar-02	SBI	0.849	Sterlite	0.447
Mar-03	SBI	0.822	Sterlite	0.434
Mar-04	SBI	0.795	Sterlite	0.345
Mar-05	SBI	0.739	Sterlite	0.509
Mar-06	SBI	0.747	Sterlite	0.136
Mar-07	SBI	0.742	Sterlite	0.155
Mar-08	SBI	0.754	Sterlite	0.120
Mar-09	SBI	0.752	Sterlite	0.117
Mar-10	SBI	0.773	Sterlite	0.206
Year	**Company**	**IP/PBIT**	**Company**	**IP/PBIT**
Mar-01	TATAS	0.444	Wipro	0.009
Mar-02	TATAS	0.625	Wipro	0.003

Mar-03	TATAS	0.217	Wipro	0.003
Mar-04	TATAS	0.084	Wipro	0.003
Mar-05	TATAS	0.036	Wipro	0.003
Mar-06	TATAS	0.033	Wipro	0.001
Mar-07	TATAS	0.039	Wipro	0.002
Mar-08	TATAS	0.118	Wipro	0.032
Mar-09	TATAS	0.172	Wipro	0.060
Mar-10			Wipro	0.020

Table 6: Correlation between IP/PBIT ratio and PAT for all Sample companies

Company	**Correlation**
ACC	-0.82
Bharti	-0.79
BHEL	-0.85
Cipla	0.52
DLF	-0.05
HDFCB	-0.07
HDFCL	-0.51
HH	-0.71
Hindalco	0.17
HUL	-0.48
ICICI	-0.19
ITC	-0.67
Infy	Zero debt
JSPL	-0.70
JPA	-0.68

L&T	-0.59
MM	-0.69
MSIL	0.04
NTPC	-0.27
ONGC	-0.78
RCom	0.68
Rel Infra	-0.13
RIL	-0.88
SBI	-0.63
Sterlite	-0.89
TATAM	0.61
TATAP	-0.51
TATAS	-0.72
TCS	-0.36
Wipro	0.51

Table 7: Lowest and Highest Autocorrelation with respective lags

Company	Lowest Autocorrelation	Lag	Highest Autocorrelation	Lag
ACC	.114	3	0..79	1
BHEL	0.031	3	0.529	1
Bharti	-0.034	7	0.542	1
Cipla	0.022	1	-0.202	7
DLF	0.04	7	0.492	1
HDFCB	0.017	4	0.448	1
HH	-0.014	3	0.654	1
Hindalco	-.07	3	0.558	1
HUL	0.06	6	-0.48	2

HDFCL	-0.024	2	0.452	1
ICICI	0.017	8	-0.171	4
ITC	0.015	4	0.25	1
Infy	NA	NA	NA	NA
JPA	0.001	5	-0.281	4
JSPL	-0.025	3	-0.481	2
L&T	-.142	3	0.648	1
MM	0.072	7	-0.273	3
MSIL	-0.075	3	0.497	1
NTPC	0.021	3	0.433	1
ONGC	0.05	1	-0.288	5
RCom	-0.33	2	-0.391	3
RIL	-0.018	3	-0.547	1
Rel Infra	-0.158	3	0.56	1
SBI	-0.08	3	0.661	1
Sterlite	-.073	3	0.639	1
TCS	0.032	7	0.163	2
TATAM	0.155	7	-0.409	5
TATAP	-.06	2	-0.405	3
TATAS	0.01	7	0.563	1
Wipro	0.195	2	0.675	1

Conclusions:

The given Table 3 gives very conclusive evidence in favour of interest free financing. Twenty of the 29 companies which used debt has a given a negative correlation between debt ratio and PAT indicating that when debt has reduced, their profitability has increased. Surprisingly some of the big names like Reliance Industries and Tata group companies are a part of this list.The Interest paid to operating profit ratio indicates the amount of interest paid per rupee of operating profit. *This value tends to zero as debt tends to zero.* It is high for banks because their major expense is Interest. For other companies the lower the better and it should tend to zero, as for the case of Infosys and Wipro, where, it is negligible and still they are earning good. Infosys is an ideal case to justify the concept of interest free financing by including more equity than debt (ideally zero) in the capital structure. Profit after Tax (PAT) is dependent upon the capital structure of an organization. This has been verified as per regression equation in Table 4 and their R-Squared values. Out of 29 companies for 15 companies Debt has a negative relationship with PAT when used with Equity. Even where debt has a positive relation with PAT, the coefficient of Debt is lower than Equity use of equity coefficient, Banks being some exceptions as their business is based on Debt. This evidence prescribes the less use of Debt and highlights the use of equity in capital structures, a principle which interest free financing propagates. Similar results were found by Ahmad (2012) in his study on Midcap companies in India. This concludes two points, one, it is better not to use debt in capital structure , two, use an optimum debt-equity combination can be used where debt should be less than equity. The second point goes against the fundamental principle of Islamic finance which prohibits interest based debt.

Based on the fundamental interest free philosophy of Islamic finance, it can be said the correlation between IP/PBIT ratio and PAT (Table 6) should ideally tend to negative 1. Thus Airtel, Sterlite and ACC qualify as some of the most efficient companies based on this parameter.

Autocorrelation Factors:

The data for PAT for all the companies was exposed to auto correlation test till lag 16. For almost all the companies the PAT was positively correlated with lag 1 and lag 2 and negatively correlated from lag 4 onwards till lag 8 after which it was not reported by the software .This was a common trend indicating the growth story of these companies. Even for RCom also, data for which is available only for five year, lag 2 and lag 3 reported negative correlations. The only exception is HUL for which PAT was positively correlated with lag 5 and lag 6 and then negatively correlated for lag 7 and lag 8.Debt should be used by the companies as little as possible and it should not be correlated with its lag values. This means that debt may be used as a special case but the companies should not use it because they have used it in past. Thus the autocorrelation should tend to zero. Autocorrelation for up to seven lags were calculated for the debt ratio for all the companies and the lowest autocorrelation with its lag is given in the table 7. Surprisingly the lowest autocorrelation for most of the companies was at a lag of 3 and 7 and the highest autocorrelation was at a lag of 1. This indicates that the debt use of companies is very highly dependent on its previous year which means that either there is falling trend or rising trend in companies in time period of 2-3 years. This also means that for a company, debt is not dependent on three year previous (lag) value. *It can be concluded that a company's debt is most like to be influenced by is value of immediate previous year.*

Thus, innovations could lead to debt instruments which do not have a fixed interest component which can be used selectively to find an optimum debt-equity combination. It is also immensely desired that new products are developed in the segment of equity which facilitates their replacement with debt. Such new contracts are desired. Butt and Hassan (2011) examined the valuation and accounting issues with reference to redeemable capital. They concluded that valuation on historical basis value is irrelevant and that Discounted Cash Flow (DCF) techniques are not appropriate. The given research provides a suggestion regarding the use of redeemable capital which can be used as an alternative to traditional debt. Bidabad et. al (2011) suggest the concept of an Interest free bonds as a substitute to conventional bonds. The paper achieves its objective by accepting the hypothesis that equity is much better than debt considering the growth and profitability criteria of companies operating in an interest based financial system. Since in India there is negligible structure available for interest fee financing, the companies may go for more use of equity and creating wealth for their owners as well as the nation. This will allow them to follow the concept of Islamic financing at least in principle.

References :

1. Ahmad,A. and Rahman,A.R.A. (2012), "The efficiency of Islamic and conventional commercial banks in Malaysia", *International Journal of Islamic and Middle Eastern Finance and Management*, Volume 5, Issue 3 ,pp. 241-263
2. Ahmad M.U.,(2012), "Islamic Finance-The Capital Structure and Profitability Perspective from Midcap Companies", *International Journal of Business and Economics Research* ,Volume 4, Issue 4, pp. 437-458
3. Ahmad, M.; Abdullah, N. and Al-Habshi, Syed M., (2006), "Efficiency of Islamic Banking in Malaysia : A Stochastic Frontier Approach", *Journal of Economic Cooperation Among Islamic Countries*, Volume 27,Issue 2, pp. 37-70
4. Alanazi,A.S.; Liu,B. and Forster,J.(2011), "The financial performance of Saudi Arabian IPOs", *International Journal of Islamic and Middle Eastern Finance and Management*, Volume 4, Issue 2,pp. 146-157
5. Arbouna,M. B.(2007), "The combination of contracts in Shariah: A possible mechanism for product development in Islamic banking and finance", *Thunderbird International Business Review*, Volume 49,Issue 3, pp. 341-370
6. Bashir, A.; Darrat, A. F. and Suliman, M. O.(1993), "Equity Capital, Profit Sharing Contracts and Investment: Theory and Evidence", *Journal of Business Finance & Accounting*, Volume 20,Issue 5, pp. 639-651
7. Bellalah, M. (2003), "Short and Long Term Financing and Capital Structure of Islamic Firms: A survey of the Literature", *International Journal of Finance*, Volume 15,Issue 4, pp. 2772-2787
8. Bidabad,B.;Hassan,A.;Ali,M.S.B. and Allahyarifard,M.(2011),'Interest Free Bonds Financial Innovation: A Monetary Instrument for Economy at Crisis, *Journal of Economic Cooperation and Development* ,Volume 32,Issue 1,pp.55-70

9. Butt,S.A. and Hassan,A (2011) , “Valuation and Accounting for Redeemable Corporate Capital: An Islamic Perspective ‘*Journal of Islamic Economics, Banking and Finance*, Volume 7,Issue 4,pp. 65-78
10. Chapra, M. U. (2000), “Is it necessary to have Islamic economics?”, *Journal of Socio-Economics*, Volume 29, Issue 1, pp. 21-37
11. Chong, B.S. and Liu, M.H. (2009), “Islamic banking: Interest-free or interest-based?”, *Pacific-Basin Finance Journal*,Volume 17,Issue 1, pp. 125-144
12. Esty, B.C. (2000), “The Equate Project: An Introduction to Islamic Project Finance”, *Journal of Project Finance*,Volume 5,Issue 4, pp. 7-21
13. Farhang , H.M.(2009), “The concept of capital before and after industrial revolution: Islamic versus market economy”, *Chinese Business Review,Volume* 8,Issue 9,pp. 44-49
14. Jafri, S.H.A. and Margolis, L.S.(1999), “The Treatment of Usury in the Holy Scriptures”, *Thunderbird International Business Review*, Volume 41,Issue 4/5,pp. 371-380
15. Jobst,A. ; Kunzel,P.; Mills,P. and Amadou Sy (2008), “Islamic bond issuance: what sovereign debt managers need to know”, *International Journal of Islamic and Middle Eastern Finance and Management*, Volume 1, Issue 4,pp.330-344
16. Looney, Robert E.(1982), “Saudia Arabia's Islamic Growth Model”, *Journal of Economic Issues*,Volume 16,Issue 2, pp. 458-459
17. Modi, V. (2007), “Writing the Rules”, *Harvard International Review*, Volume 29,Issue 1, pp. 38-41
18. Rosly, S.A. and Bakar, M.A.A. (2003), “Performance of Islamic and mainstream banks in Malaysia”, *International Journal of Social Economics*, Volume 30,Issue 11/12, pp.1249-1265

19. Shubber,K. and Alzafiri,E., (2008) , "Cost of capital of Islamic banking institutions: an empirical study of a special case", *International Journal of Islamic and Middle Eastern Finance and Management* , Volume 1, Issue 1 ,pp.52-68
20. Siddiqi, MN., (1967), *Banking without interest (Urdu)*, Lahore: Islamic publications